

RAKESH BHAGAT

+91 6289996175 | rakeshbhagat333@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL

Backend: Node.js, Express.js, RESTful APIs, WebSockets, Microservices, Webhooks (HMAC verification), Razorpay, Better Auth, FastAPI (familiar)

Frontend: React.js, Next.js (App Router, Server Actions), HTML5, CSS3, Tailwind CSS, Responsive Design

Databases: PostgreSQL, Supabase, MongoDB, Prisma ORM, Row-Level Security (RLS), Query Optimization, Indexing

Cloud & DevOps: AWS (EC2, S3), Cloudflare R2, PM2, Docker, NGINX, Certbot/SSL, CI/CD Pipelines, Vercel, Git, GitHub

AI/ML: Gemini AI API (LLM integration), Python ML Microservices (TF-IDF, Cosine Similarity), RAG Concepts, Vector-Based Search

CS Fundamentals: DBMS, Computer Networks, OS, System Design

PROFESSIONAL EXPERIENCE

Full Stack Developer, Crescentia One

December 2025 - Present

- Migrated a client's production website (ProfitMaster) off company-owned infrastructure onto the client's own accounts with zero customer-facing downtime during cutover - transferred the **GitHub** repository, moved the **Supabase PostgreSQL** database, copied **Cloudflare R2** object storage via **rclone**, transferred the **profitmaster.in** domain, and redeployed the **Next.js** app as a **Cloudflare Workers** on the client's account.
- Diagnosed slow cross-page navigation in production by indentifying **Vercel↔Supabase** region mismatch from response headers (~250 ms round-trips per query), then cut perceived load time by adding App-Router streaming skeletons and replacing per-page **auth.getUser()** network calls with cookie-only session reads across **9 routes**.
- Owned **Leopardol**, the company's e-commerce platform, as sole full-stack engineer across **3 services** (API, admin panel, storefront) from system design through production deployment.
- Migrated client infrastructure from **WordPress** to **Next.js** on **Cloudflare Workers**, re-engineering the database layer and SSR pipeline, and integrating a **Razorpay** payment gateway for event registrations that processed **Rs 3,00,000** in transactions from a single event.

KEY PROJECTS

Call AI - Real-Time Voice Agent for Banking (Python, FastAPI, Next.js, Gemini Live API, Twilio)

[Live Code](#)

- Built a real-time voice AI agent on Gemini's native-audio Live API that **verifies callers** by mobile number and date of birth, answers loan and deposit queries through **function-calling tools** backed by account data, and **escalates to a human** agent when appropriate.
- Designed a transport-agnostic call-handling core so the same agent, tools, and escalation logic serve both a browser client and live phone calls via Twilio Media Streams, **converting between 8kHz -law telephony audio and Gemini's 16/24kHz PCM** with a stateful resampler.
- Benchmarked three Gemini Live models on end-to-end **response latency (3.5s down to 1.1s)** to select the production model; deployed the FastAPI backend on a Hetzner VPS behind NGINX/PM2 and the Next.js frontend on Vercel.

SketchyDraw - Real-Time Collaborative Canvas (Node.js, PostgreSQL, WebSockets, AWS EC2, NGINX)

[Live](#)

- Engineered scalable **Node.js** backend with WebSocket-based real-time synchronization enabling multi-user collaborative drawing with **sub-100ms latency** and state consistency across all connected clients.
- Designed normalized **PostgreSQL** schema with **Prisma ORM** for type-safe CRUD operations, supporting user authentication, session management, and persistent canvas data storage.
- Configured production deployment on **AWS EC2** with NGINX reverse proxy and SSL certificates, achieving **99.9% uptime** with zero-downtime updates.

Leopardol - Full-Stack E-Commerce Platform (Next.js, Express.js, PostgreSQL, Cloudflare R2, Razorpay, AWS EC2) [Live](#)

- Architected a full-stack e-commerce platform as **3** independently deployed services - a **Node.js/Express** REST API, a **Next.js** admin panel, and a **Next.js** storefront - backed by a **20-model PostgreSQL** schema via **Prisma** and **Better Auth** RBAC, exposing **55 REST endpoints** across **14 route modules** for catalog, cart, checkout, and shipping.
- Built the checkout and payments flow on **Razorpay** with dual **HMAC-SHA256** signature verification (client callback and webhook) feeding one shared fulfillment path, using **decimal.js** for stock validation, coupon discounts, and shipping-threshold pricing to remove floating-point rounding as a source of order-total mismatches.
- Replaced server-proxied image uploads with direct-to-bucket **Cloudflare R2** storage via **AWS SDK v3** presigned **PutObject** URLs (**300s** expiry), taking the backend off the upload path for every product image.
- Deployed the API and storefront to **AWS EC2** under **PM2** process management behind an **NGINX** reverse proxy with **Certbot**-issued TLS certificates, and shipped the admin panel independently on **Vercel**, decoupling internal-tooling releases from the customer-facing deploy pipeline.

EDUCATION

Guru Ghasidas Vishwavidyalaya (Central University)

Bachelor of Technology in Electronics and Communication Engineering | CGPA: 8.65/10.0

Bilaspur, Chhattisgarh

Nov 2022 - June 2026